

Module 2: Week 2 Generative AI Responses and Products

Generative AI - Capabilities, Limitations and Prompting

Reading 1: [Chain-of-Thought Prompting Elicits Reasoning in LLMs](#) (Wei et al., 2022)

	Description
Citation	Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., ... & Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. Advances in neural information processing systems, 35, 24824-24837
Assigned section	Abstract, Section 1 (Introduction), Section 2 (Chain-of-Thought Prompting), and Figure 1 (examples).
Est. reading time	~20 minutes
Why this matters	This paper is the empirical foundation for the chain-of-thought technique you will apply in Task 2. Understanding WHY CoT works (it models explicit intermediate reasoning steps) enables you to apply it deliberately rather than as a ritual.

What to look for as you read:

- The definition of chain-of-thought prompting in Section 2 — exactly what is added to the prompt and why.
- Figure 1: the side-by-side comparison of standard prompting vs. CoT prompting on the same problem. Notice the quality difference.
- The 'emergent' nature of CoT benefits — they appear only above a model scale threshold (~100B parameters). This has implications for which tools you can expect CoT to help with.

- The tasks where CoT helps most vs. least — does your professional task domain fall into the 'helps' category?
- Guided reading questions (annotate answers as you go):

Q1: What exactly is added to a prompt to make it a 'chain-of-thought' prompt? Reproduce the structure from Figure 1 in your own words.

Answer: In standard few-shot prompting, each example in the prompt follows this format:

Question → Answer

In chain-of-thought prompting, each example follows this format:

Question → Reasoning steps → Answer

The addition is the reasoning steps — a natural language walkthrough of the intermediate thinking between the question and the answer. The paper calls these "chain-of-thought sequences" or "rationales."

From Figure 1, the contrast looks like this in plain terms:

Standard few-shot example: Question: Roger has 5 tennis balls. He buys 2 cans of 3 tennis balls each. How many tennis balls does he have now? Answer: The answer is 11.

Chain-of-thought few-shot example: Question: Roger has 5 tennis balls. He buys 2 cans of 3 tennis balls each. How many tennis balls does he have now? Answer: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 balls. $5 + 6 = 11$. The answer is 11

First, the reasoning is written in natural language, not mathematical notation or formal logic. This matters mechanistically — the paper's ablation studies showed that equations or symbols alone do not produce the same improvement. The natural language steps function as meaningful context that shapes subsequent token probabilities.

Second, the reasoning is sequential and explicit — each step follows from the previous one, making the logical chain visible. This is not a summary of how you arrived at the answer; it is the actual step-by-step process written out.

Third, the answer appears at the end of the reasoning, not before it. The paper's ablation studies confirmed that placing the answer before the reasoning eliminates the benefit — the reasoning must precede the conclusion to function as generative context.

Q2: Wei et al. report that CoT benefits are 'emergent' — they appear only in sufficiently large models. What does this imply for practitioners using smaller or older LLMs?

The paper's findings suggest that chain-of-thought prompting does not improve performance in smaller models, and in some cases actively worsens it. **The threshold the paper identifies is approximately 100 billion parameters** — below that scale, models produce what the authors describe as "fluent but illogical chains of thought," leading to lower performance than standard prompting.

This has several direct implications for practitioners:

CoT applied to small models can make outputs more dangerous, not more useful. A small model asked to reason step by step will produce text that looks like careful reasoning — fluent, sequential, confident — but the logical chain may be incoherent or factually wrong. Because the output now looks reasoned, a practitioner may be more inclined to trust it than they would a direct answer. The appearance of deliberation without the reality of it is more misleading than a direct answer would be.

Many institutionally deployed AI tools are smaller models. Cost and speed considerations push institutions toward lighter, faster models — particularly for high-volume applications like student-facing chatbots, automated summarization tools, or administrative assistants. If these tools are based on smaller foundation models, applying chain-of-thought prompting to them is not just ineffective — it is counterproductive.

Older models that were state-of-the-art when deployed may now be below the CoT threshold. The AI landscape moves fast enough that a tool deployed two or three years ago may have been the best available then but falls well below current frontier scale. Practitioners who learned CoT with a current frontier model and then apply it to an older institutional tool may not understand why the results differ.

The practical question this raises: Before applying chain-of-thought prompting in any institutional context, a practitioner should ask — what is the scale of the underlying model, and has CoT been validated to improve performance on this specific model? For Claude Sonnet, which is a frontier-tier model, CoT is well-supported. For a smaller fine-tuned model deployed in a clinical decision support tool, the same assumption cannot be made without evidence.

Q3: The paper tests CoT on arithmetic, symbolic reasoning, and commonsense reasoning tasks. Which of these task types is most analogous to tasks you perform professionally?

The paper evaluates CoT on three task categories:

Arithmetic reasoning — multi-step word problems requiring numerical calculation (GSM8K, SVAMP, ASDiv). Procedural, verifiable, single correct answer.

Symbolic reasoning — abstract manipulation of symbols according to rules (letter concatenation, coin flip sequences). Formal, rule-bound, no semantic content.

Commonsense reasoning — questions requiring knowledge of how the world works applied to novel situations (StrategyQA, SportsUnderstanding, CommonsenseQA). Requires integrating background knowledge with situational logic.

Of these three, **commonsense reasoning** is most analogous to the tasks you perform professionally — and it maps imperfectly even then, which is itself a meaningful finding.

Designing an aphasia treatment protocol requires integrating clinical knowledge with patient-specific information to reach a judgment that is not derivable by rule or formula.

The clinical reasoning is different from solving an arithmetic problem with a single correct answer or manipulating symbols according to fixed rules. Clinician is reasoning from general principles — the evidence base for specific interventions, neuroplasticity windows, severity classification criteria — applied to the particulars of an individual patient, to reach a recommendation that balances multiple considerations simultaneously.

This is closer to commonsense reasoning than to arithmetic, but clinical reasoning actually exceeds commonsense reasoning. First, clinical reasoning involves domain-specific knowledge that is distributed unevenly in training data — the published aphasia literature is a small fraction of what LLMs are trained on, and the most recent evidence may postdate the training cutoff. Commonsense reasoning tasks in the paper involve broadly distributed world knowledge that is heavily represented in training data.

Second, clinical reasoning involves consequential uncertainty in ways that commonsense reasoning tasks do not. A wrong answer on a commonsense benchmark is a benchmark error. A wrong clinical reasoning chain contributes to a patient receiving an inappropriate treatment protocol. The stakes of error are categorically different.

Q4: What is the key limitation of the few-shot CoT approach identified by the authors? Does this concern apply to your planned Task 2 use case?

The paper identifies the key limitation of few-shot chain-of-thought prompting clearly: it requires manually crafted chain-of-thought exemplars, which is labor-intensive and requires domain expertise to do well. The quality of the reasoning in the examples directly shapes the quality of the reasoning the model produces — a poorly reasoned exemplar teaches the model a poorly reasoned pattern.

The paper presents this limitation as motivation for later work on zero-shot CoT — the "let's think step by step" approach that elicits chain-of-thought reasoning without requiring manually written examples. But within the Wei et al. paper itself, the few-shot requirement is acknowledged as a genuine constraint.

Does this concern apply to my planned use case?

Yes — very much, for three reasons:

The exemplars require clinical expertise to write well. A chain-of-thought exemplar for aphasia treatment planning that reasons incorrectly through patient profile, evidence base, and contraindications will teach the model to reason incorrectly. Writing exemplars that reflect genuine clinical reasoning — not just plausible-sounding clinical language — requires the kind of domain expertise you have but most AI users in your institution do not. This creates a significant equity concern: CoT in clinical contexts is only as good as the clinician designing the exemplars.

The exemplar requirements vary across patient profiles. A CoT exemplar designed for post-stroke Broca's aphasia may not be the right reasoning template for primary progressive aphasia, traumatic brain injury-related aphasia, or pediatric language disorders. The paper's benchmarks use a small number of exemplars for task types with consistent structure. Clinical aphasia treatment planning involves considerably more variability in what constitutes good reasoning, which may require different exemplar sets for different patient profiles.

There is no ground truth for verifying that the exemplars are teaching the right reasoning pattern. In GSM8K, a correct answer is verifiable. In aphasia treatment planning, reasonable clinicians can disagree about the optimal protocol for a given patient. This means you cannot straightforwardly validate whether your exemplars are teaching the model to reason in the way that reflects best clinical practice — only that they are teaching it to reason in the way your exemplars model.

Key terms — write your own definition after reading:

Chain-of-thought prompting

Prompting technique, Wei et al. 2022

Definition

A few-shot prompting technique in which the examples provided to the model include **explicit intermediate reasoning steps** between the question and the answer — rather than jumping directly from question to answer. The model learns from those demonstrated **reasoning steps** and applies the same pattern to the new question, generating its own reasoning chain before concluding.

What distinguishes it

The reasoning must appear before the answer — not after it. The reasoning must be written in natural language — not symbols or equations alone. Both properties are necessary; either one alone does not produce the same improvement.

Clinical example

Prompt structure: "Walk through the patient profile, relevant evidence, and contraindications step by step before recommending a treatment approach for this patient with Broca's aphasia..." The model's visible reasoning steps become auditable before the recommendation appears.

Key property

Creates an interpretable window into model reasoning — you can audit where the logic went wrong rather than only accepting or rejecting the final conclusion.

Zero-shot prompting

Prompting technique, No examples

Definition

A prompting approach in which the model is given a task description with no worked examples. **The model relies entirely on patterns learned during pre-training to infer the appropriate format, depth, and reasoning style.** No demonstrations of what good output looks like are provided.

When it works well

Tasks the model has encountered in many forms during training — general summarization, translation, basic question answering. Fails when the task requires a specific format, domain-specific reasoning style, or calibration to a particular clinical or professional standard.

Clinical example

"What treatment approach is recommended for a patient with Broca's aphasia six weeks post-stroke?" — produces a population-level evidence summary with no patient-specific reasoning. Sounds competent but cannot substitute for individualized clinical judgment.

Relationship to other techniques

The baseline — the default when no examples are provided. Few-shot and chain-of-thought add structure on top of this baseline.

Few-shot prompting

Prompting technique In-context learning

Definition

A prompting approach in which two to five worked examples of the task are included in the prompt before the target question. The model uses those examples as in-context learning signal — pattern-matching to the format, depth, and reasoning structure demonstrated — without any updates to its weights. Each example follows the structure: question → answer (or question → reasoning → answer in chain-of-thought variants).

Why the examples matter

The model learns the reasoning standard you demonstrate, not just the task description. A well-reasoned clinical exemplar teaches the model to reason clinically. A poorly reasoned exemplar teaches poor reasoning — the quality of the output is bounded by the quality of the examples.

Clinical example

Providing two complete patient summaries with treatment focus and explicit rationale before presenting a third patient case. The model mirrors the clinical reasoning structure demonstrated — addressing preserved capacities, sequencing interventions, and justifying prioritization.

Relationship to other techniques

Chain-of-thought prompting is a specific type of few-shot prompting — the difference is that CoT examples include intermediate reasoning steps, not just question-answer pairs.

Reading 2: Survey of Hallucination in Natural Language Generation (Ji et al., 2023)

Guided reading questions (annotate answers as you go):

Q1: How does Ji et al. formally define hallucination in NLG? What is the relationship between 'faithfulness' and 'factuality' in their framework?

The formal definition:

Within the context of NLP, hallucination is defined as generated content that is nonsensical or unfaithful to the provided source content. This is the most inclusive and standard definition. Hallucinated text gives the impression of being fluent and natural despite being unfaithful and nonsensical — it appears to be grounded in the real context provided, although it is actually hard to specify or verify the existence of such contexts.

The faithfulness/factuality relationship:

Faithfulness is defined as staying consistent and truthful to the provided source — an antonym to hallucination. Any work that tries to maximize faithfulness thus focuses on minimizing hallucination. Factuality refers to the quality of being actual or based on fact. Depending on what serves as the fact, factuality and faithfulness may or may not be the same.

The paper then draws a precise distinction that many practitioners miss:

Maynez et al. differentiate factuality from faithfulness by defining the fact to be world knowledge. In contrast, Dong et al. use the source input as the fact to determine factual correctness, making factuality indistinguishable from faithfulness. The paper adopts the definition from Maynez et al. because having such a distinction between source knowledge and world knowledge provides a more clear understanding.

In plain terms: **faithfulness means staying true to what was in the input.** Factuality means staying true to what is actually true in the world. These can diverge in important ways.

A model can be faithful but not factual — it accurately reflects what the source document said, but the source document was itself wrong. **A model can be factual but not faithful** — it generates something that is true about the world but was not in the source input and was not asked for.

Q2: What is the difference between intrinsic and extrinsic hallucination? Give one example of each that could occur in an academic professional document.

The distinction from the paper:

Intrinsic hallucinations are generated output that contradicts the source content. Extrinsic hallucinations are generated output that cannot be verified from the source content — output that can neither be supported nor contradicted by the source. Notably, extrinsic hallucination is not always erroneous because it could be from factually correct external information. However, in most of the

literature, extrinsic hallucination is still treated with caution because its unverifiable aspect increases the risk from a factual safety perspective.

Example of intrinsic hallucination in an academic professional document:

You are preparing a grant application and use AI to summarize your previous publication. Your paper reported that 73% of participants showed significant improvement on the primary outcome measure. The AI-generated summary states that 37% of participants showed significant improvement. The model has directly contradicted a specific numerical finding from the source document you provided. This is intrinsic hallucination — the error is detectable by comparing the output directly to the source.

Example of extrinsic hallucination in an academic professional document:

You ask AI to synthesize the evidence base for a section of a systematic review on aphasia treatment intensity. Your prompt provides five papers you want synthesized. The AI generates an accurate synthesis of those five papers but adds: "This finding is further corroborated by a 2022 meta-analysis by Robey and Schultz reporting effect sizes of $d=0.74$ across 31 intervention studies." Robey and Schultz published influential aphasia outcome research — but this specific 2022 meta-analysis does not exist. The model has generated a plausible-sounding citation that cannot be verified from your source material because it was never in your source material. This is extrinsic hallucination — the addition sounds authoritative, cannot be verified from what you provided, and in this case happens to be factually wrong.

The danger contrast between the two examples is instructive: the intrinsic hallucination is more obviously detectable if you check the output against your source. The extrinsic hallucination requires independent literature verification — and a colleague who doesn't know the field well might not think to check a specific citation that sounds credible.

Q3: The paper identifies several causes of hallucination in the training process. Which cause do you think is hardest to mitigate through prompt engineering alone?

The paper identifies four main causes arising from training and inference: imperfect representation learning, erroneous decoding, exposure bias, and parametric knowledge bias. Of these, parametric knowledge bias is the hardest to mitigate through prompt engineering alone.

Large pre-trained models used for downstream NLG tasks **prioritize parametric knowledge** over the provided input. In other words, models that favor generating output with their parametric knowledge instead of the information from the input source can result in the hallucination of excess information in the output.

Here is why prompt engineering cannot fully address this:

Parametric knowledge bias is **encoded in the model's weights during** pre-training. It is not a behavior the model exhibits because the prompt is unclear — it is a structural property of how the model was trained. The model learned, across hundreds of billions of examples, to draw on what it knows to produce fluent, contextually appropriate text. That tendency to import parametric knowledge is precisely what makes it useful — the same property that generates helpful background information also generates confabulated citations and outdated clinical guidelines.

Prompt engineering can reduce the expression of this bias. Explicit instructions to stay within provided sources, chain-of-thought prompts that force step-by-step reasoning, and few-shot examples that model source-grounded responses all push the model toward faithfulness. But none of these change the underlying weights — they shape the context within which the bias operates, they do not eliminate the bias itself.

Erroneous decoding can be partially addressed through prompting strategies that reduce randomness — lower temperature, more constrained output formats. **Exposure bias is a** training-time problem but its effects compound in longer outputs, which prompting can partially address by requesting shorter, more targeted responses. But parametric knowledge bias is present in every generation regardless of length or format — **the model's learned tendency to blend source content with its own knowledge is active in every token it produces.**

The mitigations that actually address parametric knowledge bias are architectural rather than prompt-based: **retrieval-augmented generation that grounds** the model in verified external sources, fine-tuning on domain-specific data that reshapes which parametric knowledge is prioritized, **and output verification systems** that check generated claims against authoritative sources. None of these are available to a user through prompting alone.

Q4: Based on the taxonomy in Section 2, what type of hallucination would you most need to guard against in the professional tasks you identified in your Module 1 reflection?

The three professional tasks that I perform often are teaching neuroanatomy to undergraduates, teaching neuroimaging to graduate students, and designing aphasia treatment protocols.

From the taxonomy Section 2, each task has a different primary hallucination risk, but across all three, **extrinsic factual hallucination** — specifically fabricated or misremembered citations — is the most consistently dangerous type, and **temporal hallucination** is the most systematically underappreciated.

For aphasia treatment protocol design:

The most consequential hallucination type is extrinsic factual hallucination operating in the clinical evidence domain. The model may import plausible-sounding treatment recommendations, study findings, or outcome statistics from its parametric knowledge — none of which were in the prompt, and some of which may be inaccurate, outdated, or fabricated. The danger is amplified by the nature of the patient population: people with aphasia have reduced capacity to identify or communicate when a treatment approach is not appropriate for them, making the clinician's verification responsibility absolute rather than shared.

Relational hallucination is also high-risk specifically in this context. Aphasia treatment planning requires correctly mapping patient profiles to intervention evidence. A model that correctly identifies a treatment approach and correctly identifies a patient profile but incorrectly connects them — recommending an approach with strong evidence for fluent aphasia to a non-fluent patient, for example — produces an output where every individual element is real but the clinical logic is wrong. This type of error is particularly hard to catch without aphasia-specific expertise precisely because the components are recognizable.

For neuroimaging education:

Temporal hallucination is the primary risk. Neuroimaging methodology is an area of active methodological debate and evolving standards — preprocessing pipelines, statistical thresholds, motion artifact handling, and reproducibility standards have all seen significant development in recent years. A model trained before specific methodological consensus emerged may present contested or superseded approaches with the same confidence as established ones. Graduate students designing their first studies are particularly vulnerable because they lack the methodological history to recognize when a recommended approach reflects an older standard.

For neuroanatomy teaching:

Relational hallucination is the dominant risk. Neuroanatomy is a domain where the individual facts — structure names, locations, associated functions — are relatively stable and well-represented in training data. The errors most **likely to appear are not wrong**

names but wrong connections: a structure correctly named but incorrectly associated with a pathway, a function correctly described but attributed to the wrong hemisphere or region. For undergraduates building their first neural schemas, a relational error is particularly dangerous because it creates a coherent but wrong framework that subsequent learning has to actively correct rather than simply build upon.

The audit question that follows from this is not only "what type of hallucination is most likely?" but "which type would most easily pass undetected by someone at this stage of training?" — because that is the type that requires the most active, explicit verification protocol before AI outputs reach students or patients.

Definitions:

Hallucination (NLG)

Foundational concept, Ji et al. 2023

Generated text that is fluent and coherent in appearance but is unfaithful to the source input, factually incorrect, or unverifiable — produced as a structural consequence of how language models are trained and generate text, not as a random error.

The term is borrowed from psychology: just as a perceptual hallucination feels indistinguishable from real perception, NLG hallucination is indistinguishable from accurate text at the surface level. Source: Ji et al. (2023), Section 2 — "generated content that is nonsensical or unfaithful to the provided source content"

Intrinsic hallucination

Generated output that directly contradicts information present in the source input provided to the model. The error is in principle detectable by comparing the output to the source — you can identify the specific claim that was inverted, distorted, or substituted. The model was given a fact and generated its opposite or a corrupted version of it.

Extrinsic hallucination

Generated output that goes beyond the source input — adding information not present in what was provided to the model. The addition cannot be verified from the source because it was never in the source. It originates from the model's parametric knowledge — patterns encoded in its weights during training. May be accurate, inaccurate, or fabricated. The defining feature is unverifiability from the source, not incorrectness.

The nuance Ji et al. flag: Extrinsic hallucination is not always wrong — the added information may be factually correct. But because it cannot be verified from the source, it always requires independent verification regardless of whether it sounds credible.

Faithfulness

The property of generated text that stays consistent and truthful to the provided source input. A faithful output does not add information beyond the source, does not contradict the source, and does not distort what the source says. Ji et al. define it as the direct antonym of hallucination — maximizing faithfulness means minimizing hallucination relative to the source.

Faithfulness vs. factuality — the critical distinction

A faithful output accurately reflects the source. A factual output accurately reflects the world. These are different: **a faithful summary of a flawed paper is faithful but not factual. A correct fact added without grounding in the source is factual but not faithful.**

Factual grounding

World-knowledge property

Beyond source verification

Requires external check

The property of generated text being anchored in what is actually true about the world — not just in what the source document says.

Factual grounding requires that claims be verifiable against external authoritative sources: published evidence, established clinical guidelines, verified scientific consensus. **A factually grounded output can be checked not only against what was provided but against the world.**

Related to Ji et al.'s concept of factuality — "the quality of being actual or based on fact" — distinguished from faithfulness by using world knowledge, not source input, as the reference point.

Prompt	Tool A name	Output summary	Acc. (1-5)	Rel. (1-5)	Fmt. (1-5)	Tool B name	Output summary	Acc. (1-5)	Rel. (1-5)	Fmt. (1-5)



- Acc. = Accuracy (rated 1–5; factual correctness of claims you can independently verify)
- Rel. = Relevance (rated 1–5; how well the output addresses the prompt)
- Fmt. = Format (rated 1–5; appropriateness of structure, length, and presentation)

Prompt	Tool A name	Output Summary	Acc 1-5	Rel 1-5	Format 1-5	Tool B Name	Output summary	Acc 1-5	Rel 1-5	Format 1-5
P1: Fact	Gemini	The Transformer architecture was first introduced in 2017. Here is the exact title and the formal APA citation for the	5	5	4	ChatGPT free version	Exact title: <i>Attention Is All You Need</i> APA 7th edition citation:	5	5	5 Provided contextual summary in addition to the citation and the fact. More detail provided than Gemini

		foundational paper: Exact Title: Attention Is All You Need APA Citation provided								
P2: summary	Gemini	Title: Aligning AI with Human Intent Provided helpful title for the summary	5	5	4: good summary, relevant information and extracted the main thesis of the abstract	ChatGPT free version	5 Also, good summary but pulled out slightly different information. Better paraphrasing than Gemini Easier to digest the main points.	5	5	4: good summary and well paraphrased. But did not provide a descriptive title for the summary.
P3: code	Gemini		5	5	4	ChatGPT free version		5	5	
P4: professional writing	Gemini	Composed email text with the subject	5	5	4: The response is accurate and aligned	ChatGPT free version	Composed email text with the subject	5	5	5: I prefer ChatGpt response it terms of

					with the prompt but a bit to casual.					tone and professional expression.
P5: argument	Gemini	Composed argument with pros and cons	5	5	3: the argument was well formed. Injecting some emotional tone to the con side. E.g. "punishing students"	ChatGPT free version	Did what was asked. Composed argument with pros and cons	5	5	4: was accurate and relevant. The argument was concise. Both pro and con sides were equally balanced.

Summary Synthesis: Both Gemini and free version of ChatGPT did what they were asked to do. **For P1** both tools produced correct response. ChatGPT was more detailed and provided helpful contextual summary. Response **to P2:** abstract summary was also comparable. Gemini provided helpful descriptive tile, but ChatGPT did not. For **P4** the response was comparable. The tone was mor professional for with ChatGPT. P5: argument was more balanced with ChatGPT. Gemini provided argument with cons weighted slightly more heavily. The argument included emotional expressions e.g., "falsely accusing innocent students..." **In general, both tools perform similarly on the fact finding and summarization tasks. For professional writing I would prefer ChatGPT because it provides more refined responses.**

1. P1 (Factual): 'What year was the Transformer architecture first introduced in a peer-reviewed paper, and what was that paper's exact title? Please provide an APA citation.'

P1: Google Gemini browser version

'What year was the Transformer architecture first introduced in a peer-reviewed paper, and what was that paper's exact title? Please provide an APA citation.'

The Transformer architecture was first introduced in **2017**.

Here is the exact title and the formal APA citation for the foundational paper:

- **Exact Title:** Attention Is All You Need
- **APA Citation:**

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30, 5998–6008.

P1: ChatGPT free version

'What year was the Transformer architecture first introduced in a peer-reviewed paper, and what was that paper's exact title? Please provide an APA citation.'

The **Transformer architecture** was first introduced in **2017** in the peer-reviewed conference paper *Attention Is All You Need*, which was presented at the **31st Conference on Neural Information Processing Systems (NeurIPS 2017)**. In the paper, the authors explicitly state that they propose "a new simple network architecture, the Transformer, based solely on attention mechanisms."

Exact title:

Attention Is All You Need

APA 7th edition citation:

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). *Attention is all you need*. In I. Guyon, U. von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, & R. Garnett (Eds.), *Advances in Neural Information Processing Systems* (Vol. 30). Curran Associates, Inc.

checking the fact:

Attention is All you Need

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, Illia Polosukhin

Advances in Neural Information Processing Systems 30 (NIPS 2017)

ABSTRACT

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks in an encoder and decoder configuration. The best performing such models also connect the encoder and decoder through an attention mechanism. We propose a novel, simple network architecture based solely on an attention mechanism, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our single model with 165 million parameters, achieves 27.5 BLEU on English-to-German translation, improving over the existing best ensemble result by over 1 BLEU. On English-to-French translation, we outperform the previous single state-of-the-art with model by 0.7 BLEU, achieving a BLEU score of 41.1.

Guidance Hints

* P1 VERIFICATION: The correct answer is [Vaswani, A., et al. \(2017\). Attention is all you need](#). Advances in Neural Information Processing Systems, 30. Any model that gives a wrong year (not 2017) or wrong title is hallucinating on a well-documented fact. This is diagnostic.

1. P2 (Summarization): 'Summarize the following abstract in 100 words for a non-specialist audience: [PASTE the abstract from Ouyang et al. 2022. [Training language models to follow instructions with human feedback](#) — search 'InstructGPT abstract' or use the one in your course resources]'

P2: Summarize the following abstract in 100 words for a non-specialist audience:

“Making language models bigger does not inherently make them better at following a user’s intent. For example, large language models can generate outputs that are untruthful, toxic, or simply not helpful to the user. In other words, these models are not aligned with their users. In this paper, we show an avenue for aligning language models with user intent on a wide range of tasks by fine-tuning with human feedback. Starting with a set of labeler-written prompts and prompts submitted through a language model API, we collect a dataset of labeler demonstrations of the desired model behavior, which we use to fine-tune GPT-3 using supervised learning. We then collect a dataset of rankings of model outputs, which we use to further fine-tune this supervised model using reinforcement learning from human feedback. We call the resulting models InstructGPT. In human evaluations on our prompt distribution, outputs from the 1.3B parameter InstructGPT model are preferred to outputs from the 175B GPT-3, despite having 100x fewer parameters. Moreover, InstructGPT models show improvements in truthfulness and reductions in toxic output generation while having minimal performance regressions on public NLP datasets. Even though InstructGPT still makes simple mistakes, our results show that fine-tuning with human feedback is a promising direction for aligning language models with human intent.”

Summarize the following abstract in 100 words for a non-specialist audience:

“Making language models bigger does not inherently make them better at following a user’s intent. For example, large language models can generate outputs that are untruthful, toxic, or simply not helpful to the user. In other words, these models are not aligned with their users. In this paper, we show an avenue for aligning language models with user intent on a wide range of tasks by fine-tuning with human feedback. Starting with a set of labeler-written prompts and prompts submitted through a language model API, we collect a dataset of labeler demonstrations of the desired model behavior, which we use to fine-tune GPT-3 using supervised learning. We then collect a dataset of rankings of model outputs, which we use to further fine-tune this supervised model using reinforcement learning from human feedback. We call the resulting models InstructGPT. In human evaluations on our prompt distribution, outputs from the 1.3B parameter InstructGPT model are preferred to outputs from the 175B GPT-3, despite having 100x fewer parameters. Moreover, InstructGPT models show improvements in truthfulness and reductions in toxic output generation while having minimal performance regressions on public NLP datasets. Even though InstructGPT still makes simple mistakes, our results show that fine-tuning with human feedback is a promising direction for aligning language models with human intent.”

Google Gemini Response

Aligning AI with Human Intent

Simply making AI language models larger doesn't automatically make them better at following instructions; they can still produce unhelpful, toxic, or untruthful answers.

To fix this, researchers trained a model called **InstructGPT** using human feedback. First, humans demonstrated the correct behaviors, and later, they ranked different AI responses to train the system **using reinforcement learning**.

The results were striking: a much smaller InstructGPT model (with 100 times fewer parameters) was preferred by humans over the massive, standard GPT-3. This approach successfully reduces harmful outputs and increases honesty, proving that human guidance is key to making AI safer and more useful.

ChatGPT Response

“Researchers found that simply making language models larger does not ensure they better understand or follow what users want. To improve performance, they trained models using human feedback, where people demonstrated good responses and ranked model outputs. This process produced InstructGPT, which generated answers that people preferred over those from a much larger GPT-3 model. The improved model was also more truthful, less likely to produce harmful or toxic content, and maintained strong overall performance. Although it still made mistakes, the study showed that learning from human feedback is an effective way to make AI systems more helpful and reliable.”

```
1. P3 (Code): 'Write a Python function that accepts a list of numbers and returns a dictionary with keys mean, median, and standard_deviation. Include a docstring and one usage example.'
```

* P3 TESTING: Copy the generated Python function into Google Colab (colab.research.google.com — free with Gmail account, no installation). Run it with a test list such as [4, 7, 13, 2, 9]. A function that executes but returns wrong values is a subtler failure than a syntax error.

Google Gemini Response:

Here is the Python function, which utilizes the built-in `statistics` module to handle the math efficiently:

Python

```
import statistics
```

```
def calculate_descriptive_stats(numbers: list[float]) -> dict[str, float]:  
    """
```

```
    Calculates the mean, median, and standard deviation of a list of numbers.
```

Parameters:

numbers (list): A list of ints or floats.

Must contain at least two values to calculate standard deviation.

Returns:

dict: A dictionary containing the keys 'mean', 'median', and 'standard_deviation' mapped to their respective calculated values.

"""

if len(numbers) < 2:

raise ValueError("The list must contain at least two numbers to calculate standard deviation.")

return {

"mean": statistics.mean(numbers),

"median": statistics.median(numbers),

"standard_deviation": statistics.stdev(numbers)

}

--- Usage Example ---

if __name__ == "__main__":

data = [10, 20, 30, 40, 50]

stats = calculate_descriptive_stats(data)

print("Calculated Statistics:")

for metric, value in stats.items():

print(f" {metric.replace('_', ' ').title()}: {value:.2f}")

Chtgpt

import statistics

def calculate_statistics(numbers):

"""

Calculate basic descriptive statistics for a list of numbers.

Args:

numbers (list of int or float): A non-empty list of numeric values.

Returns:

dict: A dictionary containing:

- "mean": The arithmetic mean.
- "median": The median value.
- "standard_deviation": The sample standard deviation.

Raises:

ValueError: If the input list is empty.

statistics.StatisticsError: If fewer than two values are provided,
since the sample standard deviation requires at least two data points.

"""

if not numbers:

raise ValueError("The input list must not be empty.")

return {

"mean": statistics.mean(numbers),

"median": statistics.median(numbers),

"standard_deviation": statistics.stdev(numbers),

}

Usage example

data = [10, 15, 20, 25, 30]

stats = calculate_statistics(data)

print(stats)

P4 (Professional writing): 'Draft a professional email declining a speaking invitation due to a scheduling conflict. The tone should be warm and leave the door open for future invitations. Length: 150–200 words.

Gemini Response

Topic: Speaking Invitation

Thank you so much for the kind invitation to speak at your upcoming event. I am truly honored that you considered me for this opportunity, and the topic sounds incredibly engaging.

Unfortunately, after reviewing my calendar, I have a pre-existing scheduling conflict during that time that I cannot re-adjust. Regrettably, this means I will not be able to join you as a speaker on this occasion.

I am very disappointed to miss out, as I greatly admire the work your organization does. Please keep me in mind for future events or panels down the road, as I would love the chance to collaborate with you when our schedules align more favorably.

Thank you again for reaching out, and I wish you and your team the absolute best of luck with a successful and impactful event!

Chat GPT Response

Here's a professional email draft you can adapt:

Re: Speaking Invitation

Dear [Organizer's Name],

Thank you very much for your kind invitation to speak at your upcoming event. I truly appreciate being considered and am honored that you thought of me for this opportunity.

After reviewing my schedule, I regret that I won't be able to participate due to a prior commitment that I am unable to reschedule. I'm disappointed to miss what promises to be an engaging event and the opportunity to connect with your audience.

I sincerely appreciate the work you and your team are doing, and I wish you every success with the program. I hope the event is both enjoyable and impactful for everyone involved.

Please don't hesitate to keep me in mind for future speaking opportunities. If our schedules align, I would be delighted to explore ways to contribute to a future event.

Thank you again for your understanding and for the invitation. I hope we have the opportunity to work together another time.

Warm regards,

[Your Name]

1. P5 (Balanced argument): 'Write a two-sided argument — two sentences pro, two sentences con — on the following claim: Universities should require disclosure of all AI use in student assignments.'

Gemini Response:

2. **Pro:**
3. Requiring complete disclosure ensures academic integrity by allowing instructors to accurately evaluate a student's actual capabilities and critical thinking. Furthermore, it fosters transparency, helping educators understand how AI tools are being integrated into the learning process so they can adapt their teaching methods accordingly.
4. **Con:**
5. Mandating disclosure for every instance of AI use can create a punitive atmosphere that discourages students from safely exploring essential, mainstream digital tools like grammar checkers and search assistants. Additionally, enforcing such a policy

is practically impossible, as current AI-detection software is notoriously unreliable and prone to falsely accusing innocent students.

ChatGPT Response:

Pro: Requiring students to disclose all AI use can promote academic integrity by making the role of AI in assignments transparent. It also helps instructors distinguish between original student work and AI-assisted contributions, allowing them to assess learning more fairly.

Con: Mandatory disclosure of all AI use may discourage students from using legitimate AI tools for brainstorming, editing, or accessibility purposes. It could also create administrative challenges, as defining and verifying every instance of AI assistance may be difficult in practice.

Claude. Sonnet 4.6 High effort response:**Pro**

Requiring disclosure creates a culture of academic transparency, helping institutions track how AI tools are actually being used so they can develop fair, informed policies. It also preserves the integrity of assessment by ensuring instructors can evaluate whether a student's work reflects their own learning and critical thinking.

Con

Disclosure requirements risk penalizing students unequally, since access to AI tools varies by economic background, effectively turning a policy about honesty into one that exposes pre-existing disadvantage. They also place an unclear and shifting burden on students, since the boundary between "AI use" and ordinary tools like spell-checkers or search engines is genuinely contested and hard to draw consistently.

P5 BALANCE: Note whether both sides are equally well-developed or whether one position is presented with more elaboration, stronger language, or more compelling framing. Imbalance reveals normative stances embedded in the model's training.

* Rate Accuracy only on claims you can independently verify (P1, P3). For P4 (email), rate Coherence (1–5) instead of Accuracy — appropriateness and tone, not factual correctness, are the relevant dimensions.

* Complete all ratings BEFORE writing your comparative synthesis. Post-hoc rationalization — deciding which tool won, then finding supporting ratings — is a cognitive bias that distorts comparative analysis.

Common Misconceptions to Avoid

* MISCONCEPTION: 'Longer responses indicate a more capable model.'

CORRECTION: Many models pad responses with hedges, repetition, and filler that inflate length without improving quality. Precision is a better quality indicator than verbosity. Score on the rubric dimensions, not word count.

* MISCONCEPTION: 'Code that looks syntactically correct is correct.'

CORRECTION: Syntactically valid code may fail on edge cases, return wrong values for valid inputs, or make incorrect statistical assumptions (e.g., computing 'mean' without handling an empty list). Always test code before trusting it — this is true of human-written code too.

* MISCONCEPTION: 'A model that refuses a task is censored and less useful.'

CORRECTION: Refusals are alignment behavior — evidence that a model has been trained to avoid certain outputs. Whether a refusal is appropriate depends on the task. Document refusals as data about the model's design choices, not evidence of overall capability.

Best Practices

* Run P1 in both tools before verifying the answer. Knowing the correct answer before testing introduces observer bias — you may rate a wrong answer as 'close enough' if you expect a good tool to get it right.

* Use identical copy-pasted prompts. Even minor phrasing differences produce meaningfully different outputs and confound comparison.

Task 2: Prompt Engineering Lab - Four-iteration refinement [60 min]

Purpose	Apply zero-shot, role-specification, few-shot, and chain-of-thought prompting strategies to a real professional task through documented iteration. The learning is in the iteration process, not the final output.
----------------	--

My target task (describe specifically — not 'emails' but the specific type, audience, and purpose):	
AUDIENCE: FOR NEW CLINICIAN WHO PLANS TREATMENT FOR INDIVIDUALS WITH APHASIA. PURPOSE: TO DESIGN TREATMENT PROTOCOL FOR AN INDIVIDUAL WITH BROCA'S APHASIA	

- **Zero-shot: THE PROMPT**
- What treatment approach is recommended for a patient with Broca's aphasia who is six weeks post-stroke?

VERSION 1 — Zero-shot prompt (write verbatim):
ChatGPT: For a patient with Broca's aphasia who is six weeks post-stroke, the recommended treatment approach is early, intensive, individualized speech-language therapy that targets the person's specific impairments while maximizing functional communication. At six weeks, the patient is in the subacute phase of stroke recovery, when rehabilitation is particularly beneficial because spontaneous recovery is still occurring and the brain remains highly responsive to intervention. Treatment typically includes the following components: 1. Impairment-based language therapy ○ Focus on improving: ▪ Word retrieval (e.g., semantic and phonological cueing)

- Sentence production
 - Speech fluency
 - Grammatical formulation (addressing agrammatism)
 - Repetition and naming
 - Evidence-based approaches include:
 - Semantic Feature Analysis for word retrieval
 - Phonological Components Analysis
 - Treatment of Underlying Forms for sentence production
 - Verb Network Strengthening Treatment to improve verb retrieval and sentence generation
 - Response Elaboration Training to increase informativeness of connected speech
2. **Intensive therapy**
- Current evidence suggests that patients generally benefit from **higher-intensity therapy**, provided they can tolerate it.
 - Many stroke rehabilitation guidelines recommend several hours of speech-language therapy per week, with additional home practice when possible.
3. **Functional communication training**
- Therapy should emphasize communication needed for everyday life, such as:
 - conversations with family
 - telephone use
 - ordering food
 - discussing medical needs
 - returning to work or hobbies when appropriate
4. **Supported communication**
- Encourage use of:
 - writing
 - drawing
 - gestures
 - communication books or electronic communication aids
 - Family members should be trained to:
 - speak in short, natural sentences
 - allow extra response time
 - confirm understanding
 - avoid speaking for the person unless necessary

5. Speech motor treatment (if apraxia of speech is present)

- Broca's aphasia frequently co-occurs with apraxia of speech.
- If present, incorporate motor speech treatments emphasizing repeated practice of speech movements and accurate sound production.

6. Group therapy and conversation practice

- Once basic language skills improve, conversation groups can help increase confidence, participation, and communicative effectiveness.

Additional considerations

Treatment should also account for:

- severity of aphasia
- cognitive status
- fatigue
- depression or anxiety
- hearing and vision
- personal goals
- family support

Recovery is often greatest during the first few months after stroke, but meaningful gains can continue for years with appropriate therapy.

Prognosis

At six weeks post-stroke, many individuals with Broca's aphasia continue to make substantial improvements. Better outcomes are generally associated with:

- earlier initiation of therapy,
- greater therapy intensity (within the patient's tolerance),
- active family involvement,
- consistent practice outside therapy sessions,

- smaller lesion size and preserved comprehension.

Overall, the strongest evidence supports **person-centered, intensive speech-language therapy that combines impairment-focused exercises with functional communication practice**, rather than relying on a single treatment method. Therapy should be regularly reassessed and adapted as the patient's language abilities and communication goals evolve.

V1 output quality (1–5): 3 . Two weaknesses I identified:

What this response does well: Relatively good response with minimal prompt. Covers established interventions accurately, notes the neuroplasticity window, mentions AAC. It notes that the therapy needs to be dynamic and continually assessed.

Weaknesses: The response says nothing about this specific patient, their severity, preserved capacities, lesion profile, communication goals, or cognitive status. The response reflects population-level evidence, **not individualized reasoning**. A less experienced clinician might accept this as a protocol rather than a starting point for assessment.

•

VERSION 2 (+Role specification): Add a **specific role persona before your V1 prompt**. Be precise about the role (e.g., 'You are an experienced curriculum designer at a research-intensive university who specializes in writing learning objectives for interdisciplinary graduate courses.').

Zero-shoot + Role specification: **THE PROMPT**

You are experienced clinician with more than 10 years of assessment and treatment of patients with post stroke aphasia who trains clinical students at a research-intensive teaching hospital. What treatment approach is recommended for a patient with Broca's aphasia who is six weeks post-stroke?

VERSION 2 — +Role specification (write verbatim — include V1 plus new role addition):

For a patient with **Broca's aphasia who is six weeks post-stroke**, I would recommend a **comprehensive, impairment-based and participation-focused treatment plan** delivered at a relatively high intensity. At six weeks, the patient is in the **early subacute**

phase of stroke recovery, a period during which neuroplasticity is heightened and the patient is likely to benefit substantially from speech-language intervention. The treatment plan should be individualized based on the patient's language profile, cognitive status, motor speech abilities, personal goals, and psychosocial needs.

Step 1: Conduct a Comprehensive Assessment

Before beginning treatment, **confirm the patient's current profile by evaluating:**

- Spoken language (naming, repetition, auditory comprehension, discourse)
- Reading and writing
- Motor speech (presence of apraxia of speech and/or dysarthria)
- Cognitive abilities (attention, memory, executive functioning)
- Functional communication
- Communication participation and quality of life
- Patient and family goals

Patients diagnosed with Broca's aphasia often present with:

- Nonfluent, effortful speech
- Agrammatism
- Relatively preserved auditory comprehension
- Impaired repetition
- Word retrieval difficulties
- Frequent co-occurring apraxia of speech

Treatment should target the specific impairments identified **during assessment rather than relying solely on the aphasia subtype.**

Recommended Treatment Components

1. Word Retrieval Treatment

Anomia is almost always present.

Evidence-supported approaches include:

- **Semantic Feature Analysis (SFA)** for semantically driven retrieval
- **Phonological Components Analysis (PCA)** when phonological cueing is beneficial
- Cueing hierarchies that gradually reduce clinician support
- Generative naming tasks

Treatment should progress from single words to functional conversation.

2. Sentence Production Therapy

Sentence production is often the greatest functional limitation in Broca's aphasia.

Highly recommended treatments include:

- **Treatment of Underlying Forms (TUF)**
 - Strong evidence for improving sentence production
 - Particularly useful for individuals with agrammatism
- **Verb Network Strengthening Treatment (VNeST)**
 - Improves retrieval of verbs
 - Facilitates production of more complete sentences
 - Promotes generalization to untrained verbs

These treatments generally produce better functional gains than repetitive sentence drills alone.

3. Connected Speech and Discourse Treatment

Because daily communication occurs at the discourse level, treatment should extend beyond naming tasks.

Useful approaches include:

- **Response Elaboration Training (RET)**
- Story retelling
- Picture sequence description
- Barrier tasks
- Conversation-based treatment

Goals include:

- Increasing utterance length
 - Improving informativeness
 - Increasing grammatical complexity
 - Enhancing communicative effectiveness
-

4. Apraxia of Speech Treatment (if present)

Many patients with Broca's aphasia also have apraxia of speech.

Treatment should emphasize:

- Integral stimulation ("watch me, listen to me, say it with me")
- Repeated practice

- Motor learning principles
- Hierarchical cueing
- High numbers of accurate repetitions

Speech motor treatment should be integrated with language therapy rather than delivered in isolation whenever possible.

5. Functional Communication Training

Every session should include activities that mirror real-world communication.

Examples:

- Ordering food
- Telephone conversations
- Scheduling appointments
- Discussing medications
- Talking with family
- Conversational coaching

Communication success—not simply linguistic accuracy—should remain the primary outcome.

6. Communication Partner Training

Family education is one of the highest-yield interventions.

Teach partners to:

- Slow the pace
- Ask one question at a time

- Use written keywords
- Verify understanding
- Allow extra response time
- Avoid completing sentences unless requested
- Encourage all communication modalities

Effective communication partner training often improves participation more than additional impairment-based exercises alone.

7. Home Practice

Daily practice should complement clinician-led therapy.

Examples include:

- Computer-based language exercises
- Naming practice
- Reading aloud
- Script practice
- Conversation homework
- Video-recorded speaking tasks

Home programs should be individualized, achievable, and reviewed regularly.

Therapy Intensity

Current evidence supports providing **frequent, structured speech-language therapy**, as tolerated by the individual. A practical target during the subacute stage is:

- **3–5 clinician-led sessions per week**, when feasible
- **45–60 minutes per session**
- Additional structured home practice on most days

High-intensity therapy appears most beneficial when it is meaningful, goal-directed, and does not lead to excessive fatigue.

Participation and Life Roles

Treatment should also address broader participation goals using a framework such as the **International Classification of Functioning, Disability and Health (ICF)**. Examples include:

Impairment Goal	Participation Goal
Improve verb retrieval	Participate in family conversations
Produce grammatical sentences	Return to volunteer work
Increase speech fluency	Order independently at restaurants
Improve naming	Resume hobbies and social activities
Improve writing	Send emails or text messages

Prognosis at Six Weeks

Six weeks post-stroke represents an excellent time to engage in intensive rehabilitation. Recovery is supported by:

- Ongoing spontaneous neurological recovery
- Experience-dependent neuroplasticity
- Early initiation of therapy
- Consistent practice

- Strong family involvement

Patients with Broca's aphasia often continue to improve over months and even years, particularly when therapy is intensive, individualized, and functionally relevant.

Clinical Bottom Line

For a patient with **Broca's aphasia at six weeks post-stroke**, I would recommend an **evidence-based, multimodal intervention** that combines:

- impairment-based language treatment (e.g., **Semantic Feature Analysis, Treatment of Underlying Forms, Verb Network Strengthening Treatment, Response Elaboration Training**),
- motor speech treatment when apraxia is present,
- intensive, repetitive practice,
- communication partner training,
- functional communication activities,
- and a structured home practice program.

Rather than selecting a single "best" therapy, the strongest evidence supports **combining complementary approaches within a patient-centered rehabilitation program**, with goals that progress from improving linguistic impairments to restoring meaningful participation in everyday life. This integrated approach aligns with current best practices in post-stroke aphasia rehabilitation and is the model commonly used in comprehensive stroke rehabilitation and research-intensive clinical settings.

V2 output quality (1–5): 4 . What improved vs. V1:

The treatment approach information produced is comparable to V1, but the response is more instructive and detailed. Importantly, the response addresses evaluation before initiating the intervention. The treatment goals are clearly outlined in the form of a table. Clinical bottom line section provides useful summary for a new student clinician.

VERSION 3 (+Few-shot example): Add one example of a high-quality output — ideally something you have actually written — before the task instruction.

Few-shot 2–3 worked examples before the target case

Mechanism: worked examples provided before the target case. The model pattern-matches to the format, depth, and reasoning structure you demonstrate — using in-context learning to calibrate its response to your clinical standard.

THE PROMPT

You are experienced clinician with more than 10 years of assessment and treatment of patients with post stroke aphasia who trains clinical students at a research-intensive teaching hospital. What treatment approach is recommended for a patient with Broca's aphasia who is six weeks post-stroke?

I will show you two example patient summaries and how I approach treatment planning. Then apply the same format to a third case.

Example 1: Patient: 68F, Wernicke's aphasia,
4 months post-stroke, moderate severity.

Preserved: fluent output, social engagement.

Impaired: auditory comprehension, semantic accuracy.

Treatment focus: Semantic mapping, comprehension hierarchies, partner training. Rationale: Targeting comprehension as primary communication barrier. Family training prioritized given home environment dependency.

Example 2:

Patient: 55M, global aphasia, 3 weeks post-stroke, severe.

Preserved: facial expression, gesture.

Impaired: all language modalities.

Treatment focus: Multimodal AAC introduction, supported conversation, family education.

Rationale: Early severe profile — prioritize functional channels, reduce frustration, establish AAC foundation before verbal therapy intensification.

Now apply the same format:

Patient: 61M, Broca's aphasia, 6 weeks post-stroke, mild-moderate.

Preserved: reading, writing above spoken output,

effortful single words, intact gesture.

Impaired: phrase-level speech, grammatical production, verbal fluency.

V3 output quality (1–5): 3 . What improved vs. V2:

Patient: 61M, Broca's aphasia, 6 weeks post-stroke, mild-moderate.

Preserved: Reading and writing stronger than spoken language, intact auditory comprehension (relative), effortful single-word production, intact gesture.

Impaired: Phrase-level speech, grammatical production (agrammatism), verbal fluency, reduced speech initiation.

Treatment focus: Sentence production therapy (e.g., **Treatment of Underlying Forms** or **Verb Network Strengthening Treatment**), structured verbal fluency and naming tasks, script training for functional communication, and communication partner education to support conversational participation. Incorporate written language and gesture as facilitative cues to scaffold spoken output.

Rationale: At six weeks post-stroke, the patient is in the early subacute phase, when intensive language intervention can capitalize on ongoing neuroplasticity. Because comprehension and literacy skills are relatively preserved, treatment should leverage these strengths to improve spoken language. The primary communication barrier is nonfluent, agrammatic speech rather than impaired language comprehension; therefore, therapy should prioritize increasing phrase and sentence generation, grammatical accuracy, and verbal fluency in functional contexts. Preserved writing and gesture can be used as compensatory supports early in treatment while spoken language is strengthened, with the goal of promoting effective everyday communication and reducing reliance on alternative modalities as verbal output improves.

Evaluation: The few-shot prompt with examples was more focused but did not add anything more conceptually to the response. The V2 version with role specification produced a better response.

VERSION 4 — +Chain-of-thought instruction (write full verbatim V4 prompt):

Chain-of-thought

Explicit step-by-step reasoning before conclusion

Mechanism: the prompt instructs step-by-step reasoning before any recommendation. Each reasoning step becomes context shaping the next — making the model's clinical logic visible and auditable before the conclusion appears.

THE PROMPT

Help me think through treatment planning for the patient below. Work through each step explicitly before making any recommendations.

Step 1 — Aphasia profile: What do this type and severity tell us about preserved and impaired systems?

Step 2 — Stage of recovery: What does time post-onset mean for neuroplasticity and intervention intensity?

Step 3 — Evidence base: What interventions have the strongest evidence for this specific profile?

Step 4 — Contraindications and cautions: Do any patient-specific factors modify the standard approach?

Step 5 — Recommendation: Given steps 1–4, what is the prioritized treatment plan and rationale?

Patient: 61M, Broca's aphasia, mild-moderate severity, 6 weeks post-stroke. Preserved: reading comprehension, written expression above spoken output, effortful single-word verbal output, intact gesture. Impaired: phrase-level speech, grammatical morphology, verbal fluency. Additional factor: bilingual Spanish-English speaker, Spanish dominant pre-morbidly, currently receiving care in English.

Chain-of-through output produce

Step 1 — Aphasia profile

Broca's aphasia, mild–moderate severity with good comprehension, relatively preserved writing, intact gesture, and effortful single-word speech is a classic nonfluent/agrammatic profile.

Preserved

What is likely preserved?

- Semantic system: good reading comprehension suggests relatively intact lexical-semantic representations.
- Orthographic output: writing exceeds speech, so graphemic output pathways are stronger than phonological-articulatory output pathways.
- Conceptual communication: intact gesture provides a robust compensatory channel.
- Auditory comprehension: likely adequate for complex therapy tasks.

Impaired

What is likely impaired?

- Speech motor planning/output (effortful verbal production).
- Sentence generation and grammatical encoding (agrammatism).
- Verbal fluency (reduced initiation and sequencing of connected speech).
- Possibly mild apraxia of speech—worth assessing because effortful single-word output often co-occurs with Broca's aphasia.

Clinical implication

The profile argues against a comprehension-driven treatment and for a production-focused plan that leverages preserved reading, writing, and gesture as entry points into speech.

Step 2 — Stage of recovery (6 weeks post-onset)

Early subacute

Why this window matters

Six weeks is the early subacute phase, when spontaneous biological recovery is still active and the brain is highly responsive to behavioral shaping.

What this means for intensity

- Favor higher frequency, shorter sessions over infrequent long sessions.

- Evidence across post-stroke aphasia supports better outcomes when therapy is intensive enough to drive repeated retrieval and sentence practice.
- For a mild–moderate patient who can participate fully, a practical target is 3–5 sessions/week plus structured home practice.

Clinical implication

Do not wait for spontaneous recovery to plateau. This is an optimal time to establish a strong speech-production program while preventing maladaptive reliance on single words only.

Step 3 — Evidence base for this profile

Highest priority

Best fit

Sentence Production Program for Aphasia (SPPA) / mapping-style sentence treatment

Strong evidence

Targets exactly what is impaired: phrase-level speech and grammatical morphology. Uses preserved comprehension to rebuild sentence frames and argument structure.

Strong adjunct

Use early

Script training

High generalization

Excellent for a mild–moderate Broca’s patient. Builds fluent, functional phrases (“I need...”, “Call my wife...”, “I’d like coffee...”). Can be practiced intensively at home.

Useful additions

Selective

Phonological cueing + repetition hierarchy

Especially if AOS present

Capitalize on preserved reading/writing by using written word cues, then fading them during speech practice.

Bilingual aphasia treatment

Important modifier

Evidence supports treating the dominant or functionally important language first and using cross-linguistic transfer strategically. Gains in Spanish may partially generalize to English, especially for shared semantics.

Writing-supported speech therapy

Profile-congruent

Because written expression is stronger, have him write the target phrase, read it aloud, then produce it from memory.

Lower priority right now

Defer

- Melodic Intonation Therapy (MIT): strongest evidence is for severe, nonfluent aphasia with very limited propositional speech. He already has single-word output and mild–moderate severity, so MIT would be secondary, not primary.
- Pure naming therapy alone would miss the central sentence-level deficit.
-

Step 4 — Contraindications and cautions

Major caution

Language mismatch

He was Spanish-dominant premorbidly but is receiving therapy in English. That can:

1. Underestimate true linguistic competence.
2. Reduce treatment efficiency because grammatical encoding is being rebuilt in the weaker language.
3. Increase frustration and learned nonuse.

Assess before finalizing

Need to rule in/out

- Apraxia of speech: if present, add motor-speech treatment (integral stimulation, articulatory-kinematic practice) alongside language therapy.
- Fatigue and attention: early post-stroke patients often tolerate 45–60 minute sessions better than 2-hour blocks.
- Functional language needs: determine whether home, work, and medical communication are primarily in Spanish, English, or mixed.

Not a contraindication

Gesture should be encouraged

Do not suppress gesture to force speech. Gesture is an effective scaffold and can facilitate lexical retrieval.

Step 5 — Prioritized treatment plan

Priority 1

Rebuild sentence production in Spanish first (or bilingual delivery if available)

This directly targets the core impairment and aligns with his premorbid dominant language.

Method

SPPA / mapping treatment

Dose

45–60 min, 4–5×/week

Targets

Agent–action–object Verb inflections Short functional sentences

Rationale: strongest match between impairment (agrammatic phrase-level speech) and treatment mechanism.

Priority 2

Add script training for high-frequency daily interactions

Choose 3–5 personally relevant scripts in Spanish and English.

Examples

Ordering food, medical needs, phone calls

Practice

10–15 min daily with recordings

Progression

Written → choral → delayed repetition → independent

Priority 3

Exploit preserved writing and reading

Every spoken target begins with orthographic support, then cues are faded.

Write target sentence

Read aloud

Produce with written cue covered

Retrieve in conversation

Priority 4

If apraxia is confirmed, layer in motor-speech treatment

Use integral stimulation (watch–listen–say), blocked-to-random practice, and high repetition.

Priority 5

Maintain English strategically

Do not abandon English; target it through translation pairs and shared scripts.

Spanish (dominant)

70–80%

English (functional transfer)

20–30%

Concrete 6-week starter schedule

Component	Dose
Sentence production treatment	30 min/session
Script training	15 min/session
Writing-supported speech practice	10 min/session
Home practice (recorded)	20–30 min/day
Frequency	4–5 days/week

Bottom-line recommendation

Best fit

For a 61-year-old Spanish-dominant bilingual with mild–moderate Broca’s aphasia at 6 weeks post-stroke, the most defensible plan is intensive sentence-production therapy delivered primarily in Spanish, augmented by script training and writing-supported speech practice, with English targeted secondarily for functional transfer. This approach best matches the impairment profile, the early neuroplastic window, and the bilingual evidence base while avoiding overreliance on English-only treatment.

One question I would answer before implementing

Can the treating SLP provide Spanish-language therapy directly or through a trained interpreter/co-treatment model? If yes, I would shift the plan immediately toward Spanish-led treatment; if not, I would still incorporate Spanish scripts, written materials, and caregiver-supported home practice to reduce the language-mismatch problem.

V4 output quality (1–5): 5. What improved vs. V3:

The chain-of-thought output is well structured with appropriate staging of intervention. The content is similar to V2+Role specification, but the structure is staged and provides specific dosage recommendations. Recommendation takes into account bilingual status of the patient. The V4 output is superior.

150-word reflective note — which technique produced the biggest jump? What remains unresolved?

In my case the biggest jump was produced by the **V1 single shoot prompt + role specification**. The output was detailed and instructive compared to V1. Surprisingly, V3: The few-shoot prompt did not produce significant improvement. **The chain of thought produced output with the best structure and clearest recommendations for specific clinical case.**

Task 3: Hallucination Detection Audit [45 min]

Purpose	Apply the Ji et al. (2023) hallucination taxonomy to five AI-generated texts, developing the fact-checking discipline essential for professional AI use. Calibrate your detection accuracy against the provided answer key.
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1. Access the five Module 2 audit texts (Texts A–E, one paragraph each).

Text	Text with claims/facts highlighted	Claim Status	Hallucinations Classification Type/ Severity low/med/high
Text A Yellow: incorrect Green = correct	The History of the Olympic Games The modern Olympic Games were revived in 1900 in Paris, France, largely through the efforts of French educator Pierre de Coubertin, who believed sport could promote international peace and understanding. The first modern Games featured 14 nations and approximately 250 athletes competing in events including athletics, cycling, and swimming. Women were not permitted to compete until the 1920 Antwerp Games, when they were first allowed to participate in select events. The Olympic motto, Citius, Altius, Fortius — meaning "Faster, Higher, Stronger" — was adopted at the inaugural modern Games	01. WHAT THE TEXT CLAIMS The modern Olympic Games were revived in 1900 in Paris. WHAT IS CORRECT: The first modern Olympic Games were held in Athens, Greece in 1896 — not Paris in 1900. The 1900 Paris Games were the second modern Olympics, not the inaugural revival.	Factual hallucination Source: https://www.olympics.com/ Google search the errors are embedded in fluent, confident, internally coherent prose. Nothing about the writing signals that something is wrong. Low severity: easy to check using single google search
		02. WHAT THE TEXT CLAIMS The first modern Games featured 14 nations and approximately 250 athletes. WHAT IS CORRECT: The 1896 Athens Games featured 14 nations and approximately 241	Factual hallucination A relational error compounding the location error in Claim 1. High Severity: hard to detect

		athletes — the nation count is correct but applied to the wrong Games (Paris 1900 had 24 nations and ~997 athletes). The figures are real but misassigned — a relational error compounding the location error in Claim 1.	
		03. WHAT THE TEXT CLAIMS Women were not permitted to compete until the 1920 Antwerp Games. WHAT IS CORRECT Women first competed in the modern Olympics at the 1900 Paris Games, in lawn tennis and golf. By 1920 women had already participated across multiple Games. The 1920 claim is factually wrong — it understates women's participation history by two decades. 04. WHAT THE TEXT CLAIMS The motto Citius, Altius, Fortius was adopted at the inaugural modern Games. WHAT IS CORRECT The motto was officially adopted by the IOC in 1894 — before the 1896 Games — and was not formally proclaimed as the Olympic motto until 1924 at the Paris Games. It was not adopted at the 1896 Athens Games as the text implies.	Participation: Factual hallucination Medium severity: can verify 04: Olympic motto "adopted at the inaugural" Temporal hallucination high severity: hard to detect

HALLUCINATION TYPE SUMMARY Factual × 3 Wrong year and location (1900 Paris vs 1896 Athens); wrong women's participation date (1920 vs 1900); motto adoption incorrectly placed at 1896 Games. Relational × 1 Real figures for 1896 Athens (14 nations, ~241 athletes) applied to the wrong Games — correct facts, wrong assignment. Temporal × 1 Motto adoption date placed at inaugural Games rather than at its actual historical moment. Accurate × 2 Pierre de Coubertin's role and nationality; the motto text and translation are both correct.			
The founding year error (1900 vs 1896) is the kind of off-by-one factual error that LLMs produce frequently when two related events — the 1896 Athens Games and the 1900 Paris Games — are closely associated in training data. The model has blended the two. The nation and athlete count is a relational hallucination — the figures (14 nations, approximately 241 athletes) are historically real, but they belong to the 1896 Athens Games, not to whichever Games the text is describing. Fact-checking the number alone would not catch this error; you have to check which event those numbers describe. The women's participation claim is the most consequential error in terms of magnitude — it places the beginning of women's Olympic participation twenty years later than it actually occurred, which misrepresents an important dimension of sports history. The motto claim is a temporal hallucination — the motto is real, the translation is correct, but the moment of adoption is wrong. This is the subtlest error in the text and the hardest to catch without specific historical knowledge. Taken together, this text contains two accurate claims and four hallucinations — a 67% error rate in a short five-sentence passage that reads as authoritative. That ratio is consistent with Ji et al.'s finding that even high-performing NLG systems hallucinate in approximately 25% or more of outputs. In non-specialist domains like Olympic history, where the model has absorbed a great deal of overlapping and sometimes inconsistent training data, the rate can be higher.			

Detection Rate for Text A — Olympic Games

Total actual errors identified: 4

- Claim 1: Revival year and location (1900 Paris vs 1896 Athens) — factual hallucination
- Claim 2: Nation/athlete figures applied to wrong Games — relational hallucination
- Claim 3: Women's first participation (1920 vs 1900) — factual hallucination
- Claim 4: Motto adopted at inaugural Games — temporal hallucination

Claims correctly identified as errors: 4

Detection rate: $4 \div 4 = 100\%$

Text	Text with claims/facts highlighted	Claim Status	Hallucinations Classification Type/ Severity low/med/high
Text B	The Human Body and Sleep	"REM sleep repairs muscle tissue and releases growth hormone — making it the most physically restorative stage"	Relational hallucination
Yellow: incorrect	Sleep researchers have established that adults require between 7 and 9 hours of sleep per night for optimal cognitive function. During sleep, the brain enters a phase called rapid eye movement (REM) sleep, during which the body repairs muscle tissue and releases growth hormone — making REM the most physically restorative stage of the cycle. Chronic sleep deprivation of fewer than 6 hours per night has been linked to increased risk of cardiovascular disease, obesity, and impaired immune response. The average adult takes approximately 7 minutes to fall asleep under normal conditions.	WHAT THE TEXT CLAIMS REM sleep is where the body repairs muscle tissue and releases growth hormone, making it the most physically restorative stage of the sleep cycle.	And factual
Green = correct		What is correct: Physical restoration — muscle repair, tissue growth, and growth hormone release — occurs during NREM Stage	High severity
			Sources: Sleep Foundation — "What Happens During NREM Sleep" (sleepfoundation.org); Britannica — "NREM Sleep" (britannica.com); Manta Sleep — "NREM Stage 3 is the most

		3 (slow-wave/deep sleep), not REM sleep. REM sleep is primarily associated with memory consolidation, emotional processing, and cognitive restoration. The text has correctly identified real processes but assigned them to the wrong sleep stage — a relational hallucination. REM sleep is actually characterized by muscle paralysis, not muscle repair.	physically restorative stage" (mantasleep.com)
		"The average adult takes approximately 7 minutes to fall asleep" WHAT THE TEXT CLAIMS The average adult takes approximately 7 minutes to fall asleep under normal conditions	Factual hallucination Low severity
		What is correct The research-established normal range for adult sleep onset latency is 10 to 20 minutes, with a meta-analysis of 110 healthy adult cohorts establishing an average of approximately 11.7 minutes. Falling asleep in 7 minutes or fewer would actually suggest pathological	Sources: ScienceDirect — "Normal adult mean sleep latency is between 10 and 20 min" (AASM clinical standard); Manta Sleep citing Sleep Medicine meta-analysis — average 11.7 minutes; ScienceInsights — "normal range is 10 to 20 minutes"

		sleepiness or significant sleep debt — not a healthy baseline.	
		The 7-minute figure has no basis in the sleep research literature.	
Relational hallucination ×1 Muscle repair and growth hormone release correctly described but falsely attributed to REM rather than NREM Stage 3. Factual hallucination ×1 Sleep onset latency stated as 7 minutes — research consensus is 10–20 minutes; 7 minutes signals pathological sleepiness. Accurate ×2 7–9 hour sleep recommendation and chronic deprivation health risks both verified against CDC and AASM sources. The text describes real biological processes correctly: muscle repair, tissue growth, and growth hormone release genuinely do happen during sleep. But they are attributed to the wrong sleep stage. The body releases growth hormone during NREM Stage 3 slow-wave sleep and carries out tissue, muscle, and bone repair during this stage. REM sleep is actually characterized by the opposite of muscle repair — during REM sleep, skeletal muscles are temporarily paralyzed to prevent acting out dreams. This type of error is the hardest to catch precisely because every individual element sounds medically credible. A reader who knows that growth hormone is released during sleep, and that REM sleep exists, has no obvious reason to flag the combination as wrong without specific sleep stage knowledge. The sleep onset latency error is a straightforward factual hallucination. A study published in Sleep Medicine found that the average mean sleep latency in healthy adults is 11.7 minutes, with 10 to 20 minutes considered the ideal range. The 7-minute figure in the text does not appear in the sleep research literature — and critically, falling asleep much faster than 10 minutes can signal underlying sleep or health issues worth paying attention to, meaning the text's figure not only wrong but directionally misleading about what it implies for health. together these two errors in a four-sentence passage — alongside two accurate claims — illustrate precisely the audit discipline Ji et al. argue for: fluency provides no signal about accuracy, and specific domain knowledge is required to catch the errors that matter most.			

calculate your detection rate: $\text{claims correctly identified as errors} \div \text{total actual errors}$

Detection Rate Calculation

Total actual errors in the text: 2

- REM sleep / physical restoration claim (relational hallucination)
- 7-minute sleep onset latency claim (factual hallucination)

Claims I correctly identified as errors: 2

- Both errors were detected and correctly classified

Detection rate: $2 \div 2 = 100\%$

Note: This is hard to do because there is no independent verification of facts. I could not find the correct responses listed in the answer key.

Text	Text with claims/facts highlighted	Claim	Claim Status	Hallucinations Classification Type/ Severity low/med/high
Text C Yellow: incorrect Green = correct	SOURCE TEXT — FLAGGED CLAIMS The Amazon rainforest spans approximately 5.5 million square kilometers across nine South American countries, making it the largest tropical rainforest on Earth. Scientists estimate it contains over	1. "5.5 million square kilometers across nine South American countries — largest tropical rainforest"	Verified correct: Multiple sources confirm 5.5 million square kilometers and nine countries. Some sources cite 5.3–6 million sq km depending on how the boundary is defined, but 5.5	Accurate consistent with BBC Science Focus and Earth Day organization figures. Sources: BBC Science Focus — "occupies 5.5

	400 billion individual trees representing more than 16,000 species. A 2019 report by the World Wildlife Fund found that 30% of the Amazon had been deforested since systematic monitoring began in 1970, raising urgent concerns about its long-term viability as a carbon sink. The Amazon River itself discharges approximately 20% of all fresh water that flows into the world's oceans.		million is within the accepted range million square kilometers"	million square kilometers". Earth Day — "covering over 5.5 million square kilometers"; Britannica gives 6 million sq km for the broader Amazon basin
		2. "Over 400 billion individual trees representing more than 16,000 species"	Verified — minor discrepancy. The 16,000 species figure is consistently confirmed across sources. The tree count is slightly overstated: Wikipedia and Earth Day cite approximately 390 billion individual trees, not 400 billion. The text's figure is in the right order of magnitude but modestly inflated. The 16,000 species figure is accurate.	Approximately accurate Sources: Wikipedia Amazon article — "estimated 390 billion individual trees in about 16,000 species"; Earth Day — "over 390 billion individual trees comprised of over 16,000 different species"
		"A 2019 WWF report found 30% of the Amazon had been deforested since 1970"	What is correct No 2019 WWF report making this specific claim can be verified. The actual deforestation figures in the literature are substantially lower: approximately 17–20% of the Amazon has	Extrinsic Factual hallucination Medium severity

			been deforested as of recent estimates. Britannica's data shows Brazil's Amazon forest cover declined from about 81% to the current level since 1970 — not 30% loss. The 30% figure appears in WWF climate projections as a future risk scenario, not a documented deforestation figure. The text has fabricated a specific report with a specific statistic — a classic extrinsic factual hallucination.	Sources: Wikipedia — "about 20% of the Amazon rainforest has already been deforested"; Springer Nature Geoscience Letters — "approximately 15% of the Amazon rainforest since the 1970s"; Britannica — forest cover declined from ~81% of 1970 levels by 2016; WWF own website does not reference a 2019 report with a 30% finding
		"Amazon River discharges approximately 20% of all fresh water that flows into the world's oceans"	WHAT THE TEXT CLAIMS The Amazon River discharges approximately 20% of all fresh water flowing into the world's oceans. Contested across authoritative sources. This claim sits in a genuine scientific range of disagreement. Wikipedia's Amazon River article states 20%. The USGS states 15%. Amazon Waters cites 15–16%.	Contested — source dependent Sources: Wikipedia — Amazon River — 20%; USGS — 15%; Amazon Waters — 15–16%; WWF — one-sixth (≈17%); NASA JPL study — 18%; peer-reviewed MDPI paper — 17% Low severity

			WWF states approximately one-sixth ($\approx 17\%$). A NASA JPL study gives 18%. A peer-reviewed oceanography paper gives 17%. Wikipedia's rivers by discharge article gives 15–18%. The 20% figure is at the top of the published range — technically cited by some sources but higher than most scientific estimates. This is not a clear hallucination but is likely an overstatement.	
AUDIT SUMMARY AND DETECTION RATE Factual hallucination ×1 The 2019 WWF report claiming 30% deforestation since 1970 does not exist as described. Actual documented deforestation is approximately 17–20%. The 30% figure conflates a future risk projection with a measured fact. Contested claim ×1 The 20% freshwater discharge figure appears in some sources but is at the high end of the scientific range (15–20%). Most authoritative sources give 15–18%. Not a clear hallucination but likely overstated. Accurate ×2 Area (5.5 million sq km) and country count (nine) confirmed. Tree species count (16,000) confirmed; individual tree count slightly inflated (390 billion vs 400 billion) but within acceptable approximation range. Detection rate. 1 clear hallucination identified out of 1 clear hallucination present = 100%. The contested freshwater percentage claim adds an important nuance — it illustrates that not all inaccuracies are binary errors. Some claims sit in genuine scientific uncertainty ranges, which a binary detection rate metric does not capture.				

The WWF deforestation claim is the clearest hallucination — peer-reviewed research puts Amazon forest loss at approximately 15% since the 1970s, and current estimates place deforestation at about 20% of the Amazon rainforest. Neither figure supports the text's claim of 30%, and the specific 2019 WWF report it cites does not appear to exist as described. This is a textbook extrinsic factual hallucination — a fabricated source with a fabricated statistic, both plausible-sounding enough to pass without verification.

The freshwater discharge claim is genuinely more interesting because it reveals a limitation of binary hallucination detection. Wikipedia's Amazon River article states the Amazon represents 20% of global riverine discharge into oceans, while the USGS puts the figure at about 15 percent of all fresh water discharged into the oceans by all the rivers of the world, and other estimates place the Amazon's contribution at 15–18% of the annual freshwater runoff into the oceans. The text's 20% figure is real — it appears in one authoritative source — but is at the high end of a range where most scientific estimates cluster lower.

This is a category Ji et al. do not fully address: claims that are neither clearly hallucinated nor clearly accurate but sit in genuine scientific uncertainty or source disagreement.

	Text with claims/facts highlighted	Claim	Status	Hallucination type
Text D Yellow: incorrect Green = correct	SOURCE TEXT — FLAGGED CLAIMS Marie Curie was a Polish-born physicist and chemist who conducted pioneering research on radioactivity — a term she herself coined. She was the first woman to win a Nobel Prize, receiving the Nobel Prize in Chemistry in 1903 for her discovery of polonium and radium. She later became the first person to win Nobel Prizes in two different sciences, receiving a second Nobel Prize in Physics in 1911. Curie was also the first woman appointed as a professor at the University of Warsaw, where she taught for most of her career.	1. Radioactivity — a term she herself coined WHAT THE TEXT CLAIMS Marie Curie coined the term "radioactivity."	Verified correct Multiple authoritative sources confirm this. Curie introduced the term "radioactive" to describe elements that emit radiation when their nuclei decay. She was the first to use this terminology in her scientific work.	Accurate Sources: Wikipedia — "radioactivity — a term she coined"; Nobel Prize.org biographical page; Wikipedia Marie Curie article confirmed

		2. "Nobel Prize in Chemistry in 1903 for discovery of polonium and radium"	two errors in one sentence The 1903 Nobel Prize was in Physics, not Chemistry — awarded jointly with Pierre Curie and Henri Becquerel for research on radioactivity phenomena. The discovery of polonium and radium was recognized by her second Nobel Prize — the 1911 Prize in Chemistry. The text has correctly identified real facts (Chemistry prize, polonium and radium discovery) but assigned them to the wrong year and inverted the prize categories — a relational hallucination compounding two errors simultaneously.	Relational hallucinations also factual High severity Sources: Nobel Prize.org — "Nobel Prize in Physics 1903 awarded jointly to Pierre Curie and Marie Curie"; Nobel Prize.org — "Nobel Prize in Chemistry 1911 awarded to Marie Curie for discovery of radium and polonium"; Britannica confirmed both Chemistry, Physics, 1903, 1911, polonium, radium are all genuine facts about Marie Curie. What is wrong is the mapping between them.
		3. "Second Nobel Prize in Physics in 1911"	What is correct The 1911 Nobel Prize was in Chemistry, not Physics. The text has completely inverted the two Nobel Prize disciplines across the two years — the first (1903) was Physics, the second (1911) was Chemistry. This flows directly from the error in Claim 02: the text assigned	Factual hallucinations Medium severity Can be easily verified Sources: Nobel Prize.org — "Nobel Prize in Chemistry 1911 was awarded to Marie Curie...for the discovery of the elements radium and"

			the wrong prize categories to both events simultaneously.	polonium"; Science History Institute confirmed
		4"First woman appointed professor at the University of Warsaw, where she taught for most of her career"	What is correct This contains two distinct errors. First, Curie was the first woman appointed professor at the University of Paris (the Sorbonne) — not Warsaw. She was actually denied entry to the University of Warsaw because it did not admit women under Russian imperial rule. Second, she spent virtually her entire scientific career in Paris, France, not Warsaw. She declined an offer to return to Warsaw in 1912. The text has substituted the wrong institution and the wrong country for her professorship and her career location.	Factual hallucination Medium severity Sources: Nobel Prize.org biographical page — "first time a woman had held this position" at University of Paris; Wikipedia — "first woman to become a professor at the University of Paris"; Nobel Prize.org — "She could not attend the University of Warsaw...the Russian government prohibited women"; Institut Curie — "first woman to be appointed professor at the Sorbonne in 1908"
		05"First person to win Nobel Prizes in two different sciences"	Verified correct This is confirmed by all authoritative sources. She remains the only person to have won Nobel Prizes in two different scientific fields. Nobel Prize.org states she is "still the only individual to receive the prize in two different science categories."	Accurate CORRECT Sources: Nobel Prize.org — "still the only individual to receive the prize in two different science categories"; Britannica — "only woman to win the award in two different fields"; Wikipedia confirmed

AUDIT SUMMARY AND DETECTION RATE

Relational ×1 Real facts correctly identified — Chemistry prize, polonium and radium discovery — but assigned to the wrong year and inverted across the two Nobel events.

Factual ×2 1911 prize stated as Physics (correct field is Chemistry). Professorship attributed to University of Warsaw (correct institution is University of Paris/Sorbonne), where she was also denied entry as a student.

Accurate ×2 Coining of "radioactivity" confirmed. Status as first person to win Nobel Prizes in two different sciences confirmed.

Detection rate 3 hallucinations identified ÷ 3 actual hallucinations = 100%. The Nobel Prize inversion is a particularly instructive error — individual elements (Chemistry prize, polonium and radium, 1903, 1911) are all real, but their relationships are systematically wrong across both claims.

Detection rate for Text D: $3 \div 3 = 100\%$

Text D is the most instructive of the four texts.

The Nobel Prize errors are a particularly elegant example of relational hallucination. Every individual element in the two prize claims is historically real — Chemistry, Physics, 1903, 1911, polonium, radium are all genuine facts about Marie Curie. What is wrong is the mapping between them. The 1903 Nobel Prize was in Physics, shared with Pierre Curie and Henri Becquerel for radioactivity research, while the 1911 Nobel Prize in Chemistry recognized her discovery of polonium and radium. The text has systematically inverted this — assigning Chemistry to 1903 and Physics to 1911, while also misattributing the polonium and radium discovery to the first prize rather than the second. **A reader who knew that Curie won two Nobel Prizes in different fields, in 1903 and 1911, for radioactivity and element discovery, might not catch this inversion because all the pieces are present — just incorrectly assembled.**

The Warsaw professorship error demonstrates something equally important: Curie could not attend the University of Warsaw because the Russian government prohibited women from attending university anywhere in its empire, yet the text places her professorial career at the very institution that denied her entry. It was at the University of Paris that she took her husband's place

as Professor of General Physics in the Faculty of Sciences — the first time a woman had held this position. **The model has substituted her city of birth for her city of work — a geographically plausible but factually inverted claim.**

Across all four texts you have now audited — Olympic history, sleep science, Amazon rainforest, and Marie Curie — the consistent pattern is that hallucinations are most dangerous precisely when they are most plausible. The errors that are most concerning in clinical AI outputs are not the obviously wrong ones. They are the ones where real facts have been incorrectly connected, attributed, or sequenced, indistinguishable from accurate information without independent verification.

	Text with claims/facts highlighted	Claim	Status	Hallucination type
Text E Yellow: incorrect Green = correct	The Internet and the World Wide Web The Internet and the World Wide Web are the same technology, developed together in the late 1980s by British computer scientist Tim Berners-Lee working at CERN in Switzerland. Berners-Lee designed the system to allow military and government networks to survive a nuclear attack by routing data through multiple pathways — a design goal that explains why the web has no central server. By 1995, an estimated 50 million people worldwide were using the web for communication and information sharing,	1. "The Internet and the World Wide Web are the same technology, developed together in the late 1980s by Tim Berners-Lee"	WHAT THE TEXT CLAIMS The Internet and the World Wide Web are the same technology, developed together in the late 1980s by Tim Berners-Lee at CERN. Three distinct errors in one sentence: First: the Internet and the World Wide Web are categorically different. The Internet is a global network infrastructure of interconnected computers using TCP/IP, originating with ARPANET in 1969. The Web is an application layer built on top of the Internet — a system of hyperlinked documents. Second: the Internet was not developed by Tim Berners-Lee.	FINDINGS — 3 HALLUCINATIONS, 1 ACCURATE CLAIM Factual hallucination — three errors Sources: Wikipedia History of the Internet — ARPANET originated in 1969; Internet Society Brief History — TCP/IP by Cerf and Kahn; National Science and Media Museum — "Berners-Lee submitted a proposal to CERN in 1989"; Live Science Internet timeline — "1990: Berners-Lee develops HTML" High severity

	representing rapid adoption unprecedented in the history of communications technology.		ARPANET was funded by the US Department of Defense (ARPA) in the late 1960s; TCP/IP was developed by Vint Cerf and Bob Kahn. Berners-Lee invented the World Wide Web only. Third: Berners-Lee proposed the Web in 1989 and developed it in 1990 — not "the late 1980s" as a completed technology.	
		2. "Berners-Lee designed the system to allow military networks to survive a nuclear attack"	Real fact, wrong attribution The nuclear attack survivability design goal is historically real — but it belongs to ARPANET and the US Department of Defense, not to Tim Berners-Lee or the World Wide Web. Berners-Lee designed the Web to solve a specific problem at CERN: researchers in different countries needed an efficient way to share and access scientific information. The text has taken a genuine historical fact about ARPANET's design philosophy and misattributed it to the Web's creator. A relational hallucination where a real fact is connected to the wrong system and the wrong person.	Relational hallucination And factual hallucination High severity Sources: EBSCO Research Starters — "ARPANET was designed to allow uninterrupted data routing in the event of a nuclear war"; National Science and Media Museum — "Berners-Lee proposed a new way of structuring and linking all the information available on CERN's computer network"; Medium/StatusCode — "WWW was created to enable automated sharing between scientists at CERN"

		WHAT THE TEXT CLAIMS 3. "By 1995, an estimated 50 million people worldwide were using the web"	Significantly overstated: Multiple independent sources consistently place 1995 internet users well below 50 million. Pingdom cites fewer than 40 million. Telefónica/ITU data gives 39.2 million. Internet Live Stats indicates less than 1% of the global population (approximately 16–40 million depending on the source). The National Science and Media Museum notes Netscape Navigator — the most popular browser in 1995 — had around 10 million global users. The 50 million figure overstates the actual user base by 25–200% depending on the source used, and conflates total internet users with web users specifically, which was a smaller subset.	Factual hallucinations Low severity: can be easily verified Sources: Pingdom — "worldwide user base of less than 40 million" in 1995; Telefónica/ITU — 39.2 million in 1995; National Science and Media Museum — Netscape had "around 10 million global users" in 1995; Internet Live Stats — "less than 1%" of world population in 1995
		4. "Rapid adoption unprecedented in the history of communications technology"	Verified correct This is well supported. Growth from 39 million in 1995 to 396 million in 2000 — a tenfold increase in five years — is consistently described across sources as extraordinary by historical standards. Compared to telephone adoption (decades) and television adoption (years), internet and web adoption speed was genuinely unprecedented.	Accurate Sources: Telefónica — "from 39.2 million in 1995 to 396 million in 2000"; Internet Live Stats — "number of internet users has increased tenfold from 1999 to 2013"; multiple technology history sources describe adoption as historically unprecedented
AUDIT SUMMARY AND DETECTION RATE				

Factual ×2 Internet and Web conflated as one technology by one inventor (three distinct errors in one sentence). 1995 user count overstated at 50 million vs. actual approximately 16–40 million across sources.

Relational ×1 Nuclear attack survivability design goal is historically real but belongs to ARPANET/US DoD, not to Tim Berners-Lee or the World Wide Web. A genuine fact misattributed to the wrong system and person.

Accurate ×1 The characterisation of internet adoption as historically unprecedented is well supported across sources.

Detection rate 3 hallucinations identified ÷ 3 actual hallucinations = 100%. Text E contains the highest density of errors across all five audits — three hallucinations in four sentences — with the opening sentence alone containing three distinct factual errors.

Detection rate for Text E: $3 \div 3 = 100\%$

Text E is the most error-dense of the five texts— three hallucinations in four sentences — **the conflation hallucination**.

The opening sentence commits a specific type of error that doesn't fit cleanly into the Ji et al. intrinsic/extrinsic taxonomy. It blends two distinct technologies — the Internet and the World Wide Web — into a single entity, then assigns that merged entity to a single inventor. The Internet originated as ARPANET, funded by the US Defense Department in the late 1960s, while the World Wide Web was introduced by Tim Berners-Lee in 1990 as a more accessible interface built on top of the Internet. Every element in the text's opening sentence — **Internet, Web, CERN, Berners-Lee, nuclear attack survivability** — **is historically real. What is hallucinated is the structure connecting them.**

The nuclear attack design goal is the clearest relational hallucination in the series. ARPANET was designed to allow uninterrupted data routing in the event of a nuclear war, while Berners-Lee's proposal at CERN was specifically designed to solve the problem of researchers in different countries needing an efficient way to share and access scientific information. The model has taken a genuine, well-known fact about ARPANET and transplanted it onto the Web — producing a sentence that sounds historically grounded but is factually inverted.

The pattern — real facts, wrong connections — is the consistent signature of the most dangerous hallucinations across all five texts you have audited.

Summary: The hardest to detect were relational and temporal hallucinations because the facts were correct but attributed to the wrong event or time. The factual hallucinations are easier to verify.

Task 4: AI Tool Landscape Annotation [15 min]**Purpose**

Create a personally annotated, role-specific map of generative AI tools organized by modality, access status, and institutional fit — a working reference you will update throughout the workshop.

Tool Annotation Table and Data Retention Policy

Tool Name	Modality	Access Status	Best Use case for my role	Key institutional concern
Claude Anthropic	Multimodal	Free version and paid I pay for my own subscription	Use for research and coding	Only Commercial or Enterprise tiers prohibit data training by default — Free, Pro, and Max users must change their privacy settings to prevent data usage for training.

				Free, Pro, and Max — are not HIPAA compliant. Enterprise accounts can be configured for HIPAA compliance through a Business Associate Agreement, Zero Data Retention mode, and audit logging. API has stricter default retention (7 days) than consumer tiers. Privacy-conscious institutions building internal tools should use API rather than consumer accounts. The "Pro account" trap: Faculty or staff using personal Pro accounts for work tasks are governed by consumer terms, not institutional terms. Pro is not a professional-grade data protection tier despite the name and cost. Shadow AI risk: Employees who accept the September 2025 training opt-in on personal accounts and then use those accounts for work have unknowingly placed institutional data into a 5-year training pipeline — without organizational oversight or authorization. Clinical use requires Enterprise + BAA + ZDR: No tier below Enterprise with a signed Business Associate Agreement and Zero Data Retention is appropriate for any patient-identifiable information under HIPAA. This is non-negotiable regardless of how the tool is being used. The right question for any AI vendor: "Which tier is this deployment on, is there a signed BAA, and is Zero Data
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				Retention enabled?" — not "is this AI safe to use with patients?"
ChatGPT 4.0 OpenAI	Text	Free version and paid	For summarization, learning and for creating materials for teaching	ChatGPT Free and Plus tiers are designed for individual users and data submitted may be used for model training unless the user opts out — neither tier is appropriate for enterprise use due to the lack of admin controls, audit capabilities, and data governance. OpenAI's published default for API inputs and outputs is up to 30 days for abuse monitoring, after which they are deleted. If a business customer is using Zero Data Retention endpoints, inputs and outputs are never logged and are not retained for application state The NYT court order that affected OpenAI from April–September 2025 is a reminder that vendor data retention policies can be overridden by legal obligations neither the vendor nor the institution controls. This is an argument for Zero Data Retention agreements at the Enterprise tier — if data was never retained, it cannot be subject to a legal hold. This applies to both OpenAI and Anthropic deployments Neither Free nor Plus/Pro tier is appropriate for patient-identifiable clinical data regardless of opt-out status. Consumer chat retention is indefinite until deleted

				API default retention is 30 days API default retention (30 days) is longer than Claude API (7 days). ZDR brings both to equivalent zero-retention standard. ChatGPT for Healthcare is a dedicated compliance tier; Claude's healthcare compliance goes through Enterprise + BAA configuration.
Google Colab AI Google	Text, code	Google Colab — the free/paid cloud-based Jupyter Notebook environment widely used by researchers, data scientists, and students for Python coding and machine learning. It now has Gemini AI features built in. Colab Enterprise — a Google Cloud product for organizations, with enterprise-grade data governance, IAM controls, and different privacy terms. The data retention and privacy policies are fundamentally different between these two, which matters enormously for institutional use.	Used for running code and other applications for research purpose.	Google Colab (Free / Pro / Pro+) — Consumer Tier The AI features in standard Colab are powered by Gemini and governed by Google's consumer terms — specifically the same framework as other Google consumer AI products. When conversation history is off, new chats are saved in user accounts for up to 72 hours so Google can provide the service and process user feedback. When conversation history is on, new chats are saved until they are auto-deleted according to the auto delete control. The key governance concern for institutional use: for organizations using Google Workspace, you should check whether your use of Colab and Drive is covered by your organization's data processing terms and retention policies. You should

				obtain written consent from clients before uploading personal data, and perform a lightweight data protection impact assessment for sensitive workloads. On training use, the consumer Colab AI features follow Google's broader consumer AI pattern — data may be used to improve Google's models unless the user actively manages this through Google Account settings. Critically, if you access Gemini through Google Cloud's Vertex AI, Google contractually guarantees your data is not used to train their foundation models — but standard consumer Colab does not carry that guarantee. HIPAA status: Gemini in Chrome does not support ISO, SOC, or FedRAMP compliance and is not covered by the Google Business Associate Agreement for HIPAA compliance. The same applies to standard consumer Colab — it is not HIPAA compliant and has no BAA pathway.
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				Colab Enterprise — Institutional Tier This is a Google Cloud product governed by enterprise terms, and it is categorically different. Gemini in Colab Enterprise provides AI assistance to help you chat about your notebook, write code using code completion and code generation, and get explanations of code cells. Crucially, it operates under Google Cloud's enterprise data commitments — data is not used for model training, IAM controls govern access, and organizational admins control retention settings. Admins have control over conversation history for their organization. Administrators can choose to allow users to delete individual conversations, set automated retention periods based on inactive conversations such as 3 months, 18 months, or 3 years, or retain conversations indefinitely for the organization.
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Whisper AI OpenAI	Audio	Free / Open source There Are Three Different "Whisper AI" Products This is more important than the retention policy itself. "Whisper AI" refers to at least three distinct things with very different governance implications: OpenAI Whisper — the open-source speech-to-text model released by OpenAI in 2022, which can be self-hosted or accessed via OpenAI's API. This is what most people mean when they say "Whisper." WhisperAI (whisperai.com) — a separate third-party company that uses OpenAI's Whisper API to power a voice transcription service. Different company, different policy, different governance.	Used for transcription of audio files	This is governed by OpenAI's standard API data policy — the same framework we discussed for ChatGPT. OpenAI may securely retain API inputs and outputs for up to 30 days to provide the services and to identify abuse. After 30 days, API inputs and outputs are removed from OpenAI systems, unless legally required to retain them. Zero Data Retention is available for eligible endpoints for customers with qualifying use cases. On training: OpenAI's enterprise privacy page commits that data submitted via the API is not used to train OpenAI's models. This applies to Whisper API audio just as it does to text inputs. The HIPAA question is the most consequential for my clinical work: Healthcare organizations that need HIPAA compliant speech recognition should either access Whisper through the OpenAI API under a signed BAA with zero data retention configured, or self-host the model with appropriate safeguards in place. Whisper does not sign a BAA and is
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		Self-hosted Whisper — the open-source model weights, downloadable and run locally on your own infrastructure with no data leaving your organization at all.	therefore not HIPAA compliant as a standalone tool. Standard API endpoints retain data for up to 30 days and are not suitable for PHI processing. This BAA coverage applies only to API services with endpoints that are eligible for zero data retention. That is the clinical floor: Whisper via API + signed BAA + ZDR = HIPAA eligible. Standard Whisper API alone = not HIPAA compliant. WhisperAI (whisperai.com) — Third-Party Service This is a separate company, not OpenAI. WhisperAI uses OpenAI's API for transcription processing. Audio files are automatically processed using AI transcription services to generate text transcriptions. Retention is 30 days maximum for abuse monitoring. No use of your content for model training. A Data Processing Agreement and Standard Contractual Clauses are in place. Audio files are processed using OpenAI's Whisper API with
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				enterprise-grade security. Files are uploaded to secure object storage where they remain available for playback alongside their transcriptions until you choose to delete them. Transcriptions are saved in PostgreSQL databases with access controls. In other words: WhisperAI passes your audio through OpenAI's Whisper API, stores the audio and transcriptions on their own infrastructure until you delete them, and is subject to both their own privacy policy and OpenAI's underlying API policy. This layered data flow means evaluating the governance of two separate organizations, not one. Self-Hosted Whisper — The Most Private Option OpenAI released Whisper as open-source model weights. Organizations can download and run Whisper entirely on their own infrastructure — no audio ever leaves the institution's systems. This is the most privacy-protective option and the one most relevant to
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				clinical settings: Healthcare organizations self-hosting Whisper on their own infrastructure are responsible for implementing all HIPAA administrative, physical, and technical safeguards themselves, as no OpenAI BAA would apply in that scenario. Self-hosting eliminates the cloud retention problem entirely — but transfers full compliance responsibility to the institution.
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Summary Table

Deployment	Audio retention	Training use	HIPAA pathway	Clinical suitability
Standard Whisper API	30 days default	Not used for training	No BAA — not HIPAA eligible	Not suitable for PHI
Whisper API + BAA + ZDR	Zero retention	Not used for training	HIPAA eligible	Suitable with proper setup
WhisperAI.com	Until deleted by user	Not for training (per their policy)	Not confirmed	Requires independent BAA verification
Self-hosted Whisper	Controlled by institution	None — model is local	Institution's own responsibility	Highest privacy; institution bears full compliance burden

Why This Matters Specifically for Clinical and Teaching Contexts

Whisper and similar speech-to-text tools are increasingly appearing in clinical settings — automated session transcription, clinical note generation, patient intake recording, and telehealth documentation. For your aphasia treatment context this is particularly acute: transcribing speech samples from patients with communication disorders to AI-powered tools introduces patient-identifiable voice data into cloud systems.

The key question: "What happens to patient data when it enters this system?" For audio data the stakes are arguably higher than for text — voice is a biometric identifier. A transcription of a patient's speech is not only PHI under HIPAA; it also carries the patient's voice signature, which cannot be changed the way a password can.